

$$\begin{aligned}
 A(cu+dv) &= A(cu) + A(dv) \quad (\text{By Theorem 5.1.2}) \\
 &= c(Au) + d(Av) \quad (\text{Algebraic property (ii)}) \\
 &= c \cdot 0 + d \cdot 0 = 0 + 0 = 0 \quad (\text{Algebraic property (ii)})
 \end{aligned}$$

1.7 LINEAR INDEPENDENCE

- In this section, we will study homogeneous equations from the perspective of a vector equation.
- For instance, consider the equation

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (1)$$

This equation has a trivial solution, of course, where $x_1 = x_2 = x_3 = 0$. As in Section 1.5, the main issue is whether the trivial solution is the only one.

Definition: An ordered set of vectors $\{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation

$$x_1 v_1 + x_2 v_2 + \dots + x_p v_p = 0$$